

Problem Statement

In traditional software development, code changes are often tested manually, which can lead to delays, errors, and poor software quality. Modern software development adopts **DevOps practices** to automate testing and ensure rapid and reliable delivery.

Design and implement a simple DevOps pipeline that automatically builds and tests an application whenever code is pushed to a version control repository.

Tools & Technologies

- Git and GitHub (free account)
- GitHub Actions (CI tool)
- Python 3.x
- Pytest
- Any code editor (VS Code recommended)

Task to be Performed

1. **Create a GitHub Repository** named devops-lab-demo.
2. **Develop a simple Python application** that performs an arithmetic operation (e.g., addition).
3. **Write an automated test case** to verify the correctness of the application.
4. **Configure a Continuous Integration (CI) pipeline** using **GitHub Actions** that:
 - Automatically triggers on every code push
 - Installs required dependencies
 - Executes the test cases
 - Displays the test result (pass/fail)
5. **Demonstrate the DevOps workflow** by:
 - Making a code change
 - Pushing it to GitHub

- Observing the automated execution of the pipeline
6. **Intentionally introduce an error** in the code and push it again to observe pipeline failure.

Expected Repository Structure

```
devops-lab-demo/  
|  
├─ app.py  
├─ test_app.py  
├─ requirements.txt  
└─ .github/  
    └─ workflows/  
        └─ ci.yml
```

Sample Code Requirements

- `app.py` should contain a simple function.
- `test_app.py` should validate the function using **pytest**.
- `ci.yml` should define a GitHub Actions workflow for automated testing.
- `requirements.txt` should contain details of dependencies.

Expected Output

- Successful execution of the CI pipeline on correct code
- Failed pipeline execution when faulty code is pushed
- GitHub Actions workflow run screenshots

Sample “.yaml” file

```
name: CI Pipeline

on: [push]

jobs:
  build-and-test:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v4

      - name: Set up Python
        uses: actions/setup-python@v5
        with:
          python-version: '3.10'

      - name: Install dependencies
        run: pip install -r requirements.txt

      - name: Run tests
        run: pytest
```

Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.
2. The report should contain code(s), step-by-step procedure followed, output snapshots and any other asked details.
3. The PDF name should be in the form "StudentName_SAPID".